

COMPUTER ARCHITECTURE QUIZ

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Name: _____ ERP ID: _____

Instructions: Select the most appropriate option for each question.

1. What is the function of the "Functional Units" (FUs)?

- A. To store instruction bundles
- B. To execute instructions once operands are ready
- C. To rename architectural registers
- D. To ensure instructions commit in-order

2. Which stage follows "Writeback" in the overall pipeline breakdown provided in the diagram?

- A. Issue
- B. Execute
- C. Commit
- D. Fetch

3. Which phase is responsible for mapping architectural registers to physical registers?

- A. Fetch
- B. Register Renaming
- C. Execute
- D. Commit

4. What is the primary purpose of "Register Renaming" in a superscalar processor?

- A. To increase the speed of the Fetch stage
- B. To reduce the number of physical registers needed
- C. To eliminate false data dependencies like WAW and WAR
- D. To ensure instructions are executed in-order

5. What hardware structure is primarily responsible for recovering from exceptions?

- A. Branch Predictor
- B. Load/Store Queue
- C. Reorder Buffer (ROB)
- D. Physical Register File

6. According to the notes, when does an instruction's result become "architecturally visible"?

- A. When it finishes execution in the functional unit
- B. When it is renamed and issued
- C. Only after it commits/retires as the oldest instruction in the ROB
- D. Immediately upon being decoded

7. In the ROB table, why is instruction #5 [add x6, x4, x2] marked as "Wait on #4"?

- A. Structural hazard
- B. RAW dependency on x4 produced by instruction #4
- C. Branch misprediction
- D. Cache miss

8. Which is the starting number of physical registers in the examples?

- A. p20
- B. p21
- C. p22
- D. p10

9. Does register renaming remove True Data Dependencies (Read-After-Write)?

- A. Yes, it allows the dependent instruction to execute earlier
- B. No, true dependencies persist to ensure correct data linking between instructions
- C. Yes, by assigning the same physical register to both instructions
- D. No, because renaming only applies to Load/Store instructions

10. How does the processor handle a situation where a "lw" instruction does not find data in cache or main memory (e.g., virtual memory exception)?

- A. The entire processor stops and waits
- B. The exception is buffered in the ROB entry until commit
- C. It is immediately flushed from the pipeline
- D. It bypasses the renaming stage

11. Which component is responsible for "speculating on the next Program Counter (PC)"?

- A. Register Renamer
- B. Branch Predictors
- C. Functional Units
- D. Reorder Buffer

12. Based on the diagram below, determine the CPI:

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1.0 Points

CPI is

Pipeline Timing Diagram — Superscalar Out of Order with Register Renaming													
Issue width = 2 · 1 ALU (1-cycle) · 1 MUL (3-cycle) · 2 LSU (2-cycle, single port) · 5 stages: IF · ID · IS · EX · WB · C Forwarding enabled · Branch: NOT taken → misprediction													
#	Instruction	1	2	3	4	5	6	7	8	9	10	11	12
I1	lw x1, 0(x10)	IF	ID	IS	EX	EX	WB	C					
I2	lw x2, 4(x10)	IF	ID	IS	EX	EX	WB	C					
I3	add x4, x1, x2	IF	ID	-	IS	EX	WB	C					
I4	sw x4, 12(x10)	IF	ID	-	-	IS	EX	EX	WB	C			
I5	lw x5, 12(x10)		IF	ID	IS	EX	EX	WB		C			
I6	beq x9, x0, TARGET		IF	ID	-	IS	EX	WB		C			
I7	sw x9, 20(x10)				IF	ID	IS	EX	EX	WB			
I8	lw x9, 20(x10) [TARGET]				IF	ID	-	-	IS	EX	EX	WB	

- A. 3/6
- B. 3/3
- C. 6/6
- D. None

13. Observe the following Pipeline Timing Diagram:

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1.0 Points

How many instructions are incorrectly issued?

Pipeline Timing Diagram — Superscalar In-Order														
Issue width = 2 · 1 ALU (1-cycle) · 1 MUL (3-cycle) · 2 LSU (2-cycle, single port) · 5 stages: IF · ID · IS · EX · WB · C Forwarding enabled · Branch: NOT taken → correct prediction														
#	Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13
I1	lw x1, 0(x10)	IF	ID	IS	EX	EX	WB							
I2	lw x2, 4(x10)	IF	ID	IS	EX	EX	WB							
I3	add x4, x1, x2		IF	ID	.	.	IS	EX	WB					
I4	sw x4, 12(x10)		IF	ID	.	.	IS	EX	EX	WB				
I5	lw x5, 12(x10)			IF	ID	.	IS	EX	EX	WB				
I6	beq x9, x0, TARGET			IF	ID	IS	EX	WB						
I7	sw x9, 20(x10)				IF	ID	.	.	.	IS	EX	EX	WB	
I8	lw x9, 20(x10) [TARGET]				IF	ID	.	.	.	IS	EX	EX	WB	

How many instructions are incorrectly issued?

- A. 4
- B. 5
- C. 6
- D. None of these

14. Refer to the waiting stages indicated by question marks:

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1.0 Points

The question marks in the pipelined diagrams are sort of waiting stages. Which of the following introduce the waiting stages?

Superscalar Out-of-Order + Register Renaming — Pipeline Timing Diagram (Jumbled Instruction Order)																							
Issue width = 2 · 1 ALU (1-cycle) · 1 MUL (3-cycle) · 1 LSU (2-cycle) · Forwarding enabled · 6-stage pipeline: IF · ID · IS · EX · WB · C · Branch: NOT taken → correct prediction, 4-cycle flush penalty																							
#	Orig	Instruction	Renaming	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
J1	I6	add x3, x7, x5	x5 → p20, x7 → p21 x3 → p22	IF	ID	IS	EX	WB	C														
J2	I2	lw x2, 4(x10)	x10 → p19 x2 → p23	IF	ID	IS	EX	EX	WB	C													
J3	I4	mul x4, x3, x5	x4 → p24		IF	ID	IS	EX	EX	EX	WB	C											
J4	I5	add x6, x4, x2	x6 → p25		IF	ID	IS awaits J3		IS	EX	WB	C											
J5	I1	lw x1, 0(x10)	x1 → p27			IF	ID	IS	EX	EX	WB	?	C										
J6	I3	add x3, x1, x2	x3 → p28 (new)			IF	ID	IS awaits J5		IS	EX	WB	C										
J7	I7	beq x3, x0, TARGET	-				IF	ID	IS awaits J6			IS	EX	WB	C								
J8	I8	add x8, x3, x5	x8 → p29				IF	ID	Only 1 ALU			IS	EX	WB	C								
J9	I9	sw x8, 8(x10)	-					IF	ID	IS awaits J8			IS	EX	EX	WB	C						
J10	I10	sub x9, x3, x2	x9 → p30					IF	ID	Only 1 ALU			IS	EX	WB	?	C						

Which of the following introduce the waiting stages?

- A. Physical Registers
- B. Reorder Buffer
- C. In-commit stage
- D. Writeback stage